

Module 08

Decision Making: From Brain to Autonomous Systems

ITAI 4374: Neuroscience as a Model for AI
Houston City College • AI & Robotics Program
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"Module 07 asked how the brain chooses what to NOTICE. Module 08 asks how the brain chooses what to DO."



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PART 1

The Brain's Decision Machine

How does the brain choose what to do?

Section 1: Welcome — Where We Are in the Journey

Think about the last time you stood in front of a restaurant menu with too many options. Your eyes scanned the page (sensation, Module 05). Your memory reminded you that you loved the pasta last time (memory, Module 06). Your attention narrowed to two or three dishes (attention, Module 07). And then, somehow, you decided. You pointed at an item, told the server, and committed.

That moment of commitment — the shift from considering to doing — is what this module is about.

Every module we have studied so far has been building toward this one. The brain perceives the world (Module 05), stores what matters (Module 06), and selects what to focus on (Module 07). But all of that is preparation. The ultimate purpose of a brain is to produce action — to decide what to do next and then do it. An organism that senses beautifully, remembers perfectly, and attends flawlessly but never acts is an organism that does not survive.

Course Arc: SENSE (05) → REMEMBER (06) → ATTEND (07) → **DECIDE (08)** → ACT (09+)

WHAT YOU WILL LEARN

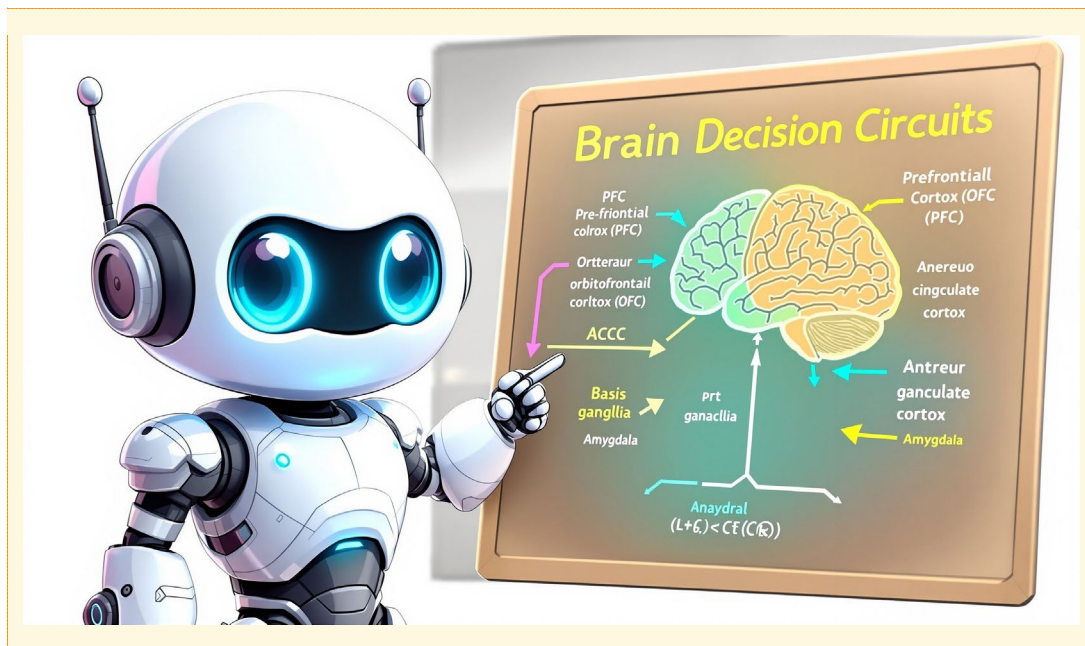
- Which brain structures work together to make decisions, and what each one contributes
- How the brain accumulates evidence before committing to a choice
- How dopamine acts as a biological error signal for learning from outcomes
- Why the brain has two decision systems — one fast and intuitive, one slow and deliberate
- Why humans make predictable irrational decisions (cognitive biases)
- How reinforcement learning in AI directly mirrors the brain's reward circuitry
- How modern AI agents use System 2 reasoning (chain-of-thought, Tree of Thoughts)
- How to think critically about when to trust AI decisions versus human judgment

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Section 2: The Decision Circuit — Who's in Charge?

Decision-making is not the job of a single brain region. It is the product of a team — a circuit of interconnected structures, each contributing something different. Think of it like a company making a strategic decision: the CEO does not act alone. The finance department runs the numbers, the risk team flags dangers, the operations team considers feasibility, and the CEO integrates all of this before committing.

The brain's decision circuit works the same way. Let us meet the team.



The Prefrontal Cortex — The CEO

We first met the prefrontal cortex (PFC) in Module 03 as the brain's executive control center. In the context of decision-making, the PFC does three critical things:

The dorsolateral PFC (dlPFC) handles working memory and abstract reasoning. When you are comparing two job offers — weighing salary against commute time against growth potential — your dorsolateral PFC is holding all those factors in mind simultaneously. In AI terms, this is the system's context window or working memory buffer.

The ventromedial PFC (vmPFC) integrates emotional information into decisions. This is where "gut feelings" get a voice in the deliberation. The importance of this region was dramatically illustrated by the case of Phineas Gage, a railroad construction foreman in Vermont. In 1848, an accidental explosion drove a three-foot iron rod through the front of Gage's skull, destroying much of his vmPFC. Remarkably, Gage survived and could still walk, talk, and reason logically. But his personality changed completely — the once responsible, well-liked foreman became impulsive, rude, and unable to make good decisions. He could analyze options rationally but

could not use emotional signals to guide his choices. His case was one of the first to show that emotions are not the enemy of good decisions — they are essential to them. We will return to this idea when we discuss the somatic marker hypothesis later in this section.

The orbitofrontal cortex (OFC) computes the subjective value of options. When you are deciding between pizza and sushi, your OFC is assigning a “value score” to each option based on your current state (How hungry are you? When did you last have pizza?). The OFC updates these values in real time — which is why food you craved five minutes ago can suddenly seem unappealing after you smell something else.

AI CONNECTION: The PFC as an AI Agent’s Planner

The three PFC subdivisions map remarkably well to components of modern AI agents:

dIPFC → Working memory / context window: Holds relevant information during multi-step reasoning.

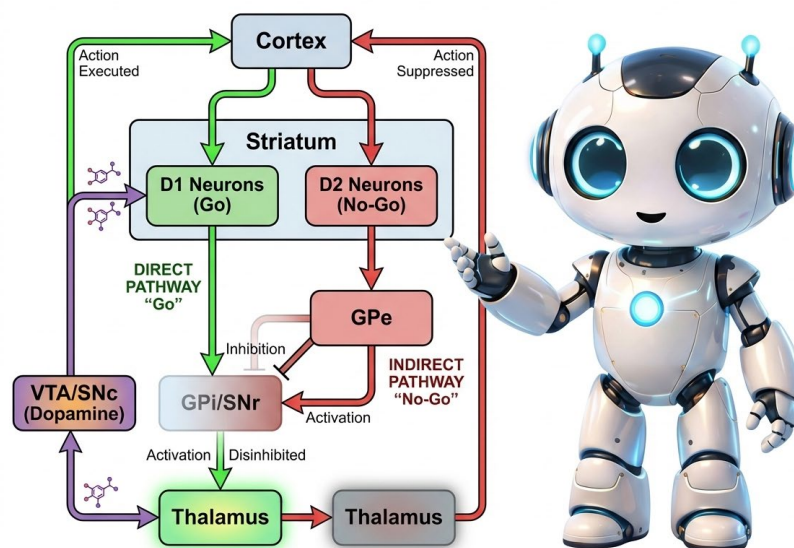
vmPFC → Reward model / value function: Evaluates how “good” an outcome feels.

OFC → Dynamic value estimation: Updates preferences based on current state, like a Q-value that changes with context.

When you build an LLM-based agent that reasons, evaluates, and plans, you are building a simplified version of the PFC.

The Basal Ganglia — The Action Selector

THE BASAL GANGLIA GO/NO-GO CIRCUIT



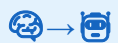
The basal ganglia are a group of structures deep in the brain that we introduced in Module 03 as the brain's actor-critic system. Now we can go deeper into how they actually select actions.

The basal ganglia work through two competing pathways:

The Direct Pathway (“Go”): When a potential action looks promising, the direct pathway releases the brake on that action, allowing it to proceed. Think of it as a green light.

The Indirect Pathway (“No-Go”): When an action should be suppressed, the indirect pathway keeps the brake engaged. Think of it as a red light.

Every moment, multiple possible actions are competing for execution. The basal ganglia resolve this competition by selectively releasing the brake on the winning action while keeping all others inhibited. Dopamine from the substantia nigra (a small structure in the brainstem that manufactures dopamine — its name means “black substance” because it contains dark pigment) modulates this competition: it strengthens the Go pathway for rewarding actions and strengthens the No-Go pathway for actions that led to bad outcomes. When the substantia nigra degenerates, as in Parkinson’s disease, patients have difficulty initiating movements — the Go pathway loses its fuel.



AI CONNECTION: Basal Ganglia = Actor-Critic Architecture

The basal ganglia implement biological reinforcement learning with an elegant architecture:

Cortex proposes candidate actions (like a policy network generating options)

Striatum evaluates them (like a critic/value network scoring each option)

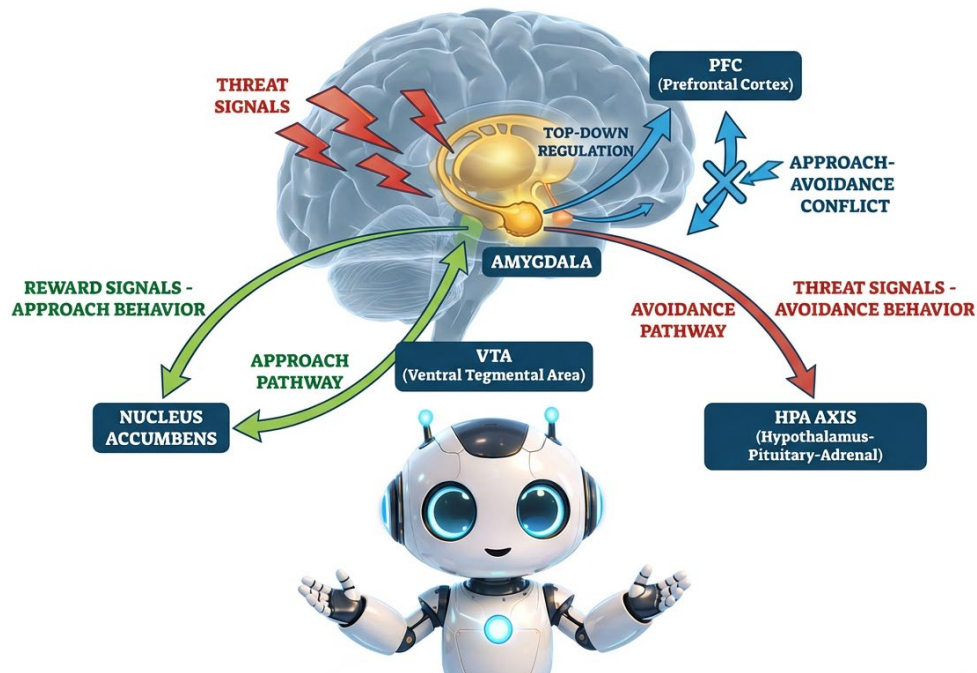
Dopamine provides the reward signal (like the TD error in Q-learning)

Globus pallidus gates the winner (like an argmax selecting the highest-scored action)

This is literally the actor-critic architecture that AI researchers developed independently. The brain had it first, by millions of years.

The Amygdala — The Risk Assessor

THE AMYGDALA IN DECISION-MAKING: APPROACH VS. AVOIDANCE



We met the amygdala in Module 03 as the brain's emotional significance detector. In decision-making, the amygdala plays a specific and crucial role: it rapidly assesses risk and threat, often before you are consciously aware of the danger.

The amygdala has bidirectional connections with the prefrontal cortex. This means emotions and rational analysis are in constant dialogue during decisions. When you feel uneasy about a deal that looks good on paper, that is your amygdala sending warning signals to your PFC. When you override your fear to give a public speech, that is your PFC sending inhibitory signals back to the amygdala.

The approach-avoidance conflict: Many decisions involve both potential reward and potential risk. Should you invest in a volatile stock? Accept a job in a new city? Try the unfamiliar dish? The amygdala and the reward system (the ventral striatum — the “reward center” in the front part of the basal ganglia that lights up when you anticipate something good) are essentially arguing with each other, and the PFC mediates. This tug-of-war is the biological basis of what psychologists call approach-avoidance conflict.

The Somatic Marker Hypothesis — Why “Gut Feelings” Matter

Neuroscientist Antonio Damasio proposed one of the most influential ideas in decision neuroscience: the somatic marker hypothesis. The word “somatic” comes from the Greek word

soma, meaning body. So a somatic marker is a body signal — a racing heart, a knot in the stomach, sweaty palms, a feeling of dread or excitement. Damasio's insight was that these body signals are not just side effects of thinking — they actively help guide your choices, often before you are consciously aware of why you prefer one option over another.

Damasio studied this with a now-famous experiment called the Iowa Gambling Task. Participants chose cards from four decks. Two decks gave big wins but even bigger occasional losses (bad decks). Two decks gave smaller wins but tiny losses (good decks). Over time, healthy participants started avoiding the bad decks — and here is the key finding: their skin conductance (a measure of how much your palms sweat, which increases when you feel anxious or excited) spiked before they reached for a bad deck, even before they could consciously explain why they were avoiding it. Their bodies “knew” before their minds did.

Patients with damage to the ventromedial PFC (like Phineas Gage, whose story we told above) could not generate these somatic markers. They understood the probabilities intellectually but kept choosing the bad decks anyway. Their logical reasoning worked fine, but without emotional body signals to guide them, their real-world decisions were catastrophically poor. This is one of the strongest pieces of evidence that pure logic alone is not enough for good decision-making — you need emotions too.



AI CONNECTION: Somatic Markers and AI Decision Systems

Current AI systems lack somatic markers entirely. They have no body, no emotional responses, no “gut feelings.” This is both a strength (AI does not panic under pressure) and a significant limitation (AI cannot use emotional shortcuts that help humans make fast, reasonably good decisions in complex situations).

RLHF (Reinforcement Learning from Human Feedback) can be seen as a crude attempt to give AI systems something like somatic markers — training the model to “feel” that some outputs are better than others based on human preferences, even when it cannot articulate the precise rules.

The Anterior Cingulate Cortex — The Conflict Monitor

The anterior cingulate cortex (ACC), which we encountered as part of the executive attention network in Module 07, plays a specialized role in decisions: it detects conflict. When two options are close in value and you are struggling to choose, your ACC fires intensely. It is essentially the brain's “this is a hard decision” alarm.

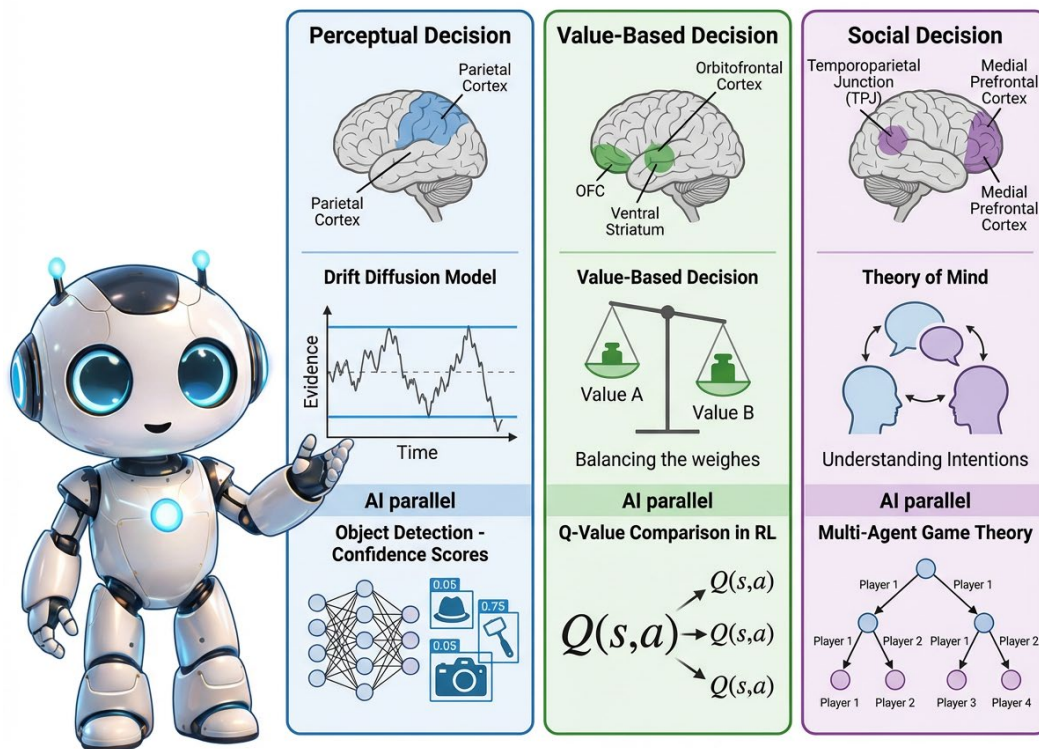
The ACC also monitors for errors. When a decision leads to a worse outcome than expected, the ACC signals that something went wrong, prompting the system to adjust. In AI terms, the ACC is like a loss function that spikes when predictions are wrong.

? KNOWLEDGE CHECK

1. A patient has damage to their orbitofrontal cortex. They can still reason logically, but they keep choosing options that are objectively worse. Why might this happen? (Hint: what does the OFC compute?)
2. The basal ganglia use a Go/No-Go system to select actions. What would happen if dopamine were depleted from this system? (Hint: think about Parkinson's disease.)

Answers are discussed in the following sections — but try to answer from what you know first.

Section 3: Three Types of Decisions



Not all decisions are the same. Neuroscientists distinguish three major categories, each relying on different brain circuits and operating under different rules.

Perceptual Decisions

A perceptual decision is based on interpreting ambiguous sensory evidence. Is that shape in the dark a person or a tree? Is that sound footsteps or rain? These decisions are fast, often unconscious, and depend on the sensory cortices (Module 05) feeding evidence to the parietal and prefrontal cortex, which accumulate it until a threshold is reached.

Key characteristic: speed-accuracy tradeoff. You can decide quickly (but risk errors) or take more time (and be more accurate). The brain adjusts this tradeoff based on context — in a life-threatening situation, speed wins; in a medical diagnosis, accuracy wins. We will formalize this in Section 4 with the drift diffusion model.

Value-Based Decisions

A value-based decision involves choosing between options with different subjective worth. Should you save money or buy new headphones? Take the safe job or the exciting startup? These decisions depend heavily on the orbitofrontal cortex (assigning value), the ventral striatum (predicting reward), and the anterior cingulate (weighing costs against benefits).

Key characteristic: subjective and context-dependent. The same option can have different values depending on your current state, recent experiences, and emotional context. A glass of water has low value after lunch but enormous value after a desert hike. This subjectivity is both a strength (flexible, adaptive) and a weakness (susceptible to biases, which we will explore in Section 7).

Social Decisions

Social decisions involve other people — trust, cooperation, competition, fairness. Should you share resources with a partner who might betray you? These decisions recruit brain regions specialized for understanding other minds:

- **Temporoparietal junction (TPJ):** Enables perspective-taking — imagining what another person is thinking or feeling
- **Medial prefrontal cortex (mPFC):** Supports mentalizing — reasoning about others' beliefs, desires, and intentions
- **Anterior insula:** Detects fairness violations — the feeling of “that’s not right” when someone cheats

Key characteristic: theory of mind. Social decisions require modeling what other agents are thinking, including what they think you are thinking. This recursive reasoning (“I think that you think that I think...”) is something humans do naturally but AI systems find extremely challenging.

Dimension	Perceptual	Value-Based	Social
What is decided?	What am I sensing?	What do I want?	What should I do given others?
Key brain region	Parietal cortex	OFC + ventral striatum	TPJ + medial PFC
Speed	Fast (ms to seconds)	Medium (seconds)	Slow (requires deliberation)
Primary challenge	Noisy/ambiguous input	Subjective preferences	Modeling other minds
AI parallel	Classifiers, object detection	Recommendation engines	Multi-agent systems, game theory

THINK ABOUT IT

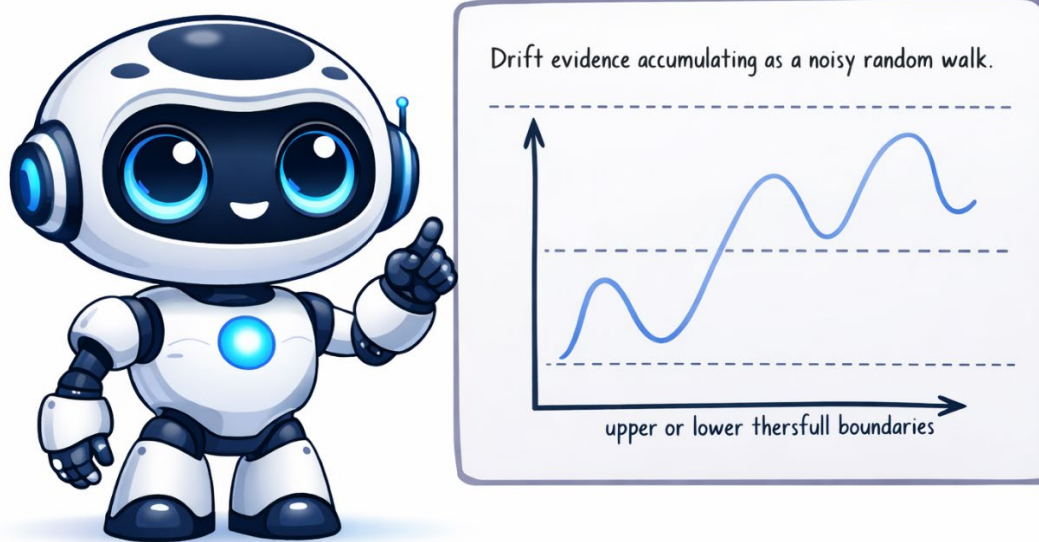
When a self-driving car decides whether a shape ahead is a pedestrian or a shadow, it is making a perceptual decision. When a recommendation algorithm suggests a movie you might like, it is making a value-based decision. When an AI negotiation agent decides whether to cooperate or compete, it is making a social decision. All three types appear in modern AI systems.

Section 4: Evidence Accumulation — How the Brain Commits to a Choice

How does the brain actually cross the line from “considering” to “deciding”? Research over the past few decades has converged on a surprisingly elegant answer: the brain accumulates evidence over time until a threshold is reached, and then it commits.

Imagine you are trying to decide if a faint sound is a doorbell or your imagination. Your auditory cortex starts collecting evidence. Each moment, tiny bits of evidence trickle in — some pointing toward “doorbell,” some toward “not doorbell.” Your brain keeps a running tally. When the evidence for one option crosses a critical threshold, you decide and act (you get up, or you dismiss it).

This process has been directly observed in neurons. Neuroscientists recording from monkey parietal and prefrontal cortex during decision tasks found that neural firing rates gradually build up over hundreds of milliseconds, ramping toward a threshold — and the moment the threshold is crossed, the monkey commits to a choice. Faster ramp = faster decision. Noisier evidence = slower ramp and more errors.



The Drift Diffusion Model

The most successful formal description of this process is the Drift Diffusion Model (DDM). Think of it as a ball rolling on a tilted table with random bumps:

- **The drift** is the tilt — how strongly the evidence points in one direction. Strong evidence = steep tilt = fast, accurate decision.
- **The noise** is the random bumps — variability in the evidence. High noise = more errors and slower decisions.
- **The threshold** is the edge of the table — how much evidence you need before committing. High threshold = slower but more accurate. Low threshold = faster but more error-prone.
- **The starting point** is where the ball begins — your prior bias. If you already expect the doorbell (you ordered pizza), the ball starts closer to the “doorbell” edge.

MATH MOMENT (Optional): The Drift Diffusion Equation

The DDM is described by: $dx = \mu \cdot dt + \sigma \cdot dW$

In plain English:

x = your running evidence tally (starts near zero)

μ = the drift rate (quality of evidence — how tilted the table is)

σ = the noise (randomness in the evidence)

dW = a random walk (each moment, a random bump up or down)

The decision is made when x reaches either the upper boundary (choose option A) or the lower boundary (choose option B). This simple equation accounts for reaction times, accuracy rates, and the speed-accuracy tradeoff observed in humans and animals.

AI CONNECTION: Evidence Accumulation in AI Systems

The drift diffusion model has direct parallels in AI:

Bayesian inference: Prior beliefs are updated with each new piece of evidence. The posterior probability drifts toward a decision boundary.

Recurrent neural networks: Hidden states accumulate information over time steps, effectively implementing a learned evidence accumulator.

Autonomous vehicles: Sensor fusion systems accumulate confidence from multiple sensors (camera, LIDAR, radar) before committing to a driving decision.

LLM token generation: At each step, the model accumulates context to select the next token, with temperature controlling the noise/threshold tradeoff.

Section 5: Dopamine and the Reward Prediction Error — The Brain's Teaching Signal

In Module 03, we introduced one of the most remarkable discoveries in neuroscience: the work of Wolfram Schultz and colleagues in 1997 showing that dopamine neurons encode **reward prediction errors** — the difference between what you expected and what actually happened. Now we can appreciate the full significance of this discovery for decision-making.

Three Scenarios

1. Better than expected (positive prediction error): You try a new restaurant and the food is amazing. Dopamine neurons fire a burst of activity. This signal strengthens the neural pathways that led to this choice, making you more likely to choose this restaurant again. In AI terms, this is a positive TD error that increases the Q-value of the chosen action.

2. Exactly as expected (no prediction error): You go to your usual restaurant and get the usual quality. Dopamine neurons show no change in firing. There is nothing new to learn — the system is already calibrated. In AI terms, the TD error is zero, and no weight update occurs.

3. Worse than expected (negative prediction error): Your favorite restaurant is terrible tonight. Dopamine neurons dip below their baseline firing rate. This suppresses the pathways that led to this choice. Next time, you might try somewhere else. In AI terms, this is a negative TD error that decreases the Q-value.

MISCONCEPTION ALERT

"Dopamine is the pleasure chemical." This is one of the most widespread misconceptions in popular science. Dopamine does not signal pleasure itself. It signals **surprise** — specifically, the mismatch between expectation and reality. Dopamine fires when something is *better than expected*, not when something feels good. A drug addict's dopamine system fires in anticipation of the drug (prediction), not during the high (consumption). The actual feeling of pleasure is more closely linked to opioid systems.

The beauty of this system is that over time, the brain learns to predict rewards accurately, so dopamine responses shift from the reward itself to the cue that predicts the reward. Consider Pavlov's famous experiment from the 1900s: Ivan Pavlov rang a bell every time he fed his dogs. Eventually, the dogs began salivating when they heard the bell alone, even with no food present. The dogs did not salivate because the bell was rewarding — they salivated because their dopamine system learned that the bell predicted food. The bell became a "reward prediction," and the dopamine response shifted from the food to the bell. This is exactly how cues (advertisements, store logos, notification sounds) trigger cravings and behaviors in everyday life.

Dopamine and Learning Rate

Dopamine does more than just signal surprise. The size of the dopamine burst controls how much the brain learns from an experience. A massive positive surprise produces a big dopamine burst, which causes big synaptic changes — equivalent to a high learning rate in AI. A mild surprise produces a small burst and small changes — a low learning rate. This is biologically elegant: you learn the most from experiences that violate your expectations the most.

KEY TAKEAWAY

Dopamine is the brain's teaching signal, not its pleasure signal. It encodes the difference between expected and actual outcomes — exactly what AI researchers call the temporal difference (TD) error. The brain invented reinforcement learning millions of years before computer scientists formalized it.

KNOWLEDGE CHECK

1. Wolfram Schultz discovered that dopamine neurons fire when an unexpected reward arrives. What happens to that dopamine response after many repetitions, when the reward becomes fully predictable?
2. You try a new coffee shop and it's amazing — better than expected. According to the reward prediction error framework, what signal does your brain generate, and how does that shape your behavior going forward?
3. If dopamine encodes prediction error (surprise), not pleasure itself, what does it mean that dopamine fires at the cue (the sound of a coffee machine) rather than at the coffee? What has the brain learned to predict?

Section 6: Exploration vs. Exploitation — The Curiosity Dilemma

You love your favorite coffee shop. The coffee is consistently good, the barista knows your order, and you never have to think about it. But a new place just opened across the street. Do you try it?

This is the exploration-exploitation dilemma, and it is one of the fundamental problems in decision-making — both for brains and for AI systems.

Exploitation means choosing what you already know works. It is reliable, efficient, and low-risk. But it means you might miss something better.

Exploration means trying something new. It is uncertain and potentially wasteful. But it is the only way to discover better options.

The brain has evolved elegant mechanisms for balancing this tradeoff, and they map directly onto strategies used in AI.

The Brain's Solution

Dopamine drives exploration. Novel, unexpected stimuli trigger dopamine release in the ventral striatum, creating a sense of curiosity and motivation to investigate. This is why new things feel exciting — your dopamine system is giving you a reward for exploring. In AI terms, this is an exploration bonus or intrinsic reward.

Habits drive exploitation. As behaviors are repeated and rewarded, control shifts from the ventral striatum (flexible, exploratory) to the dorsal striatum (automatic, habitual). Your morning routine is dorsal striatum territory — efficient and fast, but rigid. In AI terms, this is the shift from a stochastic policy to a more deterministic one as training progresses.

The prefrontal cortex mediates. The PFC can override both systems. It can force exploration when habits are failing (“this restaurant keeps disappointing me, let me try somewhere new”) or enforce exploitation when curiosity is counterproductive (“I need to finish this report, not browse the internet”).

AI CONNECTION: Exploration Strategies in RL

AI researchers have developed several strategies to balance exploration and exploitation, each inspired by different aspects of biology:

ϵ -greedy (epsilon-greedy): This is the simplest strategy. You pick a number between 0 and 1 — say 0.1 (10%). Then, 10% of the time you try a completely random action (explore), and 90% of the time you pick the best known action (exploit). It is crude but effective — like flipping a coin every tenth decision to decide whether to try the new coffee shop.

Upper Confidence Bound (UCB): Instead of random exploration, you specifically target actions you are uncertain about. If you have been to Restaurant A ten times and Restaurant B only once, UCB says to try Restaurant B again — not because it is random, but because you do not have enough information about it yet. This is smarter than random exploration.

Softmax/Boltzmann exploration: Actions with higher known values are chosen more often, but you always leave some chance of trying alternatives. It is like being 80% likely to pick your favorite restaurant, 15% likely to pick your second favorite, and 5% likely to try somewhere new. The probabilities reflect what you know so far.

Intrinsic motivation / curiosity: Give the agent a bonus reward for discovering new states. This directly mirrors the dopamine-driven curiosity system in the ventral striatum.

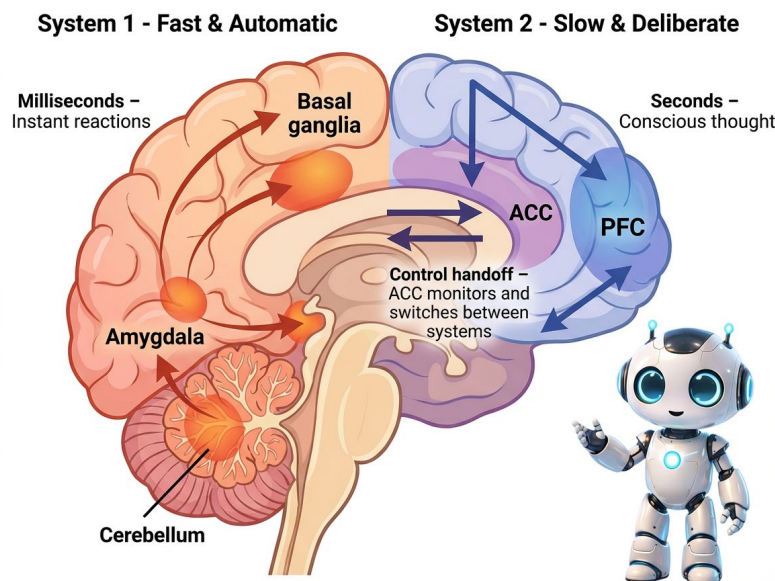
💡 THINK ABOUT IT

Social media algorithms face the exploration-exploitation dilemma constantly. Should they show you content similar to what you already like (exploitation) or introduce new topics (exploration)? Most platforms heavily exploit — creating filter bubbles. The brain, by contrast, has a built-in curiosity drive that pushes toward exploration. What would a “neuroscience-inspired” recommendation algorithm look like?

? KNOWLEDGE CHECK

1. You have been going to the same restaurant every Friday for a year. You're considering trying the new place next door. Which decision strategy are you using if you go to the new place, and which if you stay? What brain system drives each choice?
2. Social media feeds are designed to maximize engagement. Using the explore-exploit framework, explain how these algorithms can become “exploitation traps” — and what that means for the quality of information users see.
3. The prefrontal cortex can override both habit (exploitation) and novelty-seeking (exploration). Give a real-world example of a situation where a person would need to deliberately override their exploitation tendency and force exploration instead.

Section 7: System 1 vs. System 2 — Two Ways the Brain Decides



In 2011, psychologist Daniel Kahneman published *Thinking, Fast and Slow*, which popularized a framework for understanding two fundamentally different modes of human decision-making. This framework has become essential for understanding both human cognition and modern AI systems.

System 1: Fast, Intuitive, Automatic

System 1 operates quickly, effortlessly, and without conscious deliberation. It is the system that:

- Detects that one object is farther away than another
- Completes the phrase “bread and...”
- Makes you flinch when a ball flies toward your face
- Reads the emotion on someone’s face
- Drives your car on a familiar route while your mind wanders

System 1 is fast because it relies on pattern matching, learned associations, and heuristics (mental shortcuts). Neurologically, it depends heavily on the basal ganglia (habitual action selection), the amygdala (rapid threat assessment), and sensory cortices (automatic pattern recognition). It is efficient and usually correct — but it is also the source of cognitive biases, which we will explore in the next section.

System 2: Slow, Deliberate, Effortful

System 2 is engaged when you need to think carefully. It is the system that:

- Calculates 17×24 in your head
- Compares two products by reading their specifications
- Fills out a tax form
- Monitors your behavior at a formal dinner
- Navigates through an unfamiliar city using a map

System 2 is slow because it requires working memory, sustained attention, and sequential reasoning. Neurologically, it depends on the prefrontal cortex (especially the dorsolateral PFC for working memory) and the anterior cingulate cortex (for conflict monitoring). System 2 is accurate but expensive — it is metabolically costly, fatiguing, and limited by working memory capacity (~4 items, as we learned in Module 06).

Dimension	System 1	System 2
Speed	Fast (milliseconds)	Slow (seconds to minutes)
Effort	Effortless, automatic	Effortful, tiring
Awareness	Below conscious awareness	Conscious, deliberate
Capacity	Unlimited parallel	Limited (~4 items in working memory)
Basis	Pattern matching, heuristics	Logic, rules, computation
Brain regions	Basal ganglia, amygdala, sensory cortex	PFC (dlPFC), ACC
Accuracy	Usually right, but biased	More accurate, but slow and costly
AI parallel	Single forward pass (inference)	Chain-of-thought, Tree of Thoughts, ReAct

KEY TAKEAWAY

Most of your daily decisions are System 1 — automatic and unconscious. System 2 is only activated when System 1 encounters something unexpected, complex, or important. The brain defaults to the cheap system and only escalates when necessary. This is exactly the design philosophy behind modern AI agent architectures that use fast inference for routine tasks and deliberative reasoning only when needed.

Section 8: Cognitive Biases — When System 1 Gets It Wrong

System 1 is fast and efficient, but its reliance on shortcuts means it makes systematic errors — not random mistakes, but predictable patterns of irrationality. These are cognitive biases, and understanding them is critical for anyone building AI systems, because biases in human data become biases in AI models.

Anchoring

The first number you hear influences your judgment, even if it is completely irrelevant. If someone asks “Is the population of Turkey more or less than 200 million?” and then asks you to estimate the actual population, your answer will be higher than if the anchor were 20 million. This is anchoring — System 1 latches onto available numbers and adjusts insufficiently.

Workforce connection: AI pricing algorithms exploit anchoring by showing inflated “original prices” next to discounts. AI salary negotiation tools use anchoring strategically.

Loss Aversion and Prospect Theory

Daniel Kahneman and Amos Tversky discovered that losses feel roughly twice as painful as equivalent gains feel pleasurable. Losing \$100 hurts more than finding \$100 feels good. This asymmetry is called loss aversion, and it is the foundation of prospect theory — a model of how humans actually evaluate risky decisions (as opposed to how a perfectly rational agent would).

Neurologically, loss aversion involves the amygdala (threat response to potential losses) overriding the OFC's value calculations. From an evolutionary perspective, this makes sense: in a survival context, losing your food supply is fatal, while gaining extra food is nice but not essential.

Framing Effect

The same information presented differently leads to different decisions. “This surgery has a 90% survival rate” feels very different from “This surgery has a 10% death rate,” even though they are identical. System 1 responds to the emotional frame (survival vs. death) rather than computing the underlying probability.

Sunk Cost Fallacy

You continue investing in something because of what you have already spent, even when the rational choice is to stop. You sit through a terrible movie because you paid \$15, even though leaving would save two hours of your life. The \$15 is gone regardless — but System 1 cannot let go of it. AI systems do not naturally suffer from sunk cost fallacy, which is actually an advantage.

Confirmation Bias

You seek out and favor information that confirms what you already believe, while ignoring or dismissing contradicting evidence. This is deeply connected to the brain's predictive coding system from Module 02 — the brain literally expects to find confirming evidence and treats disconfirming evidence as a “prediction error” that it tries to minimize.

Workforce connection: AI systems trained on biased data exhibit a computational version of confirmation bias — they find patterns that reinforce existing distributions in the training data, even when those patterns reflect societal prejudice rather than ground truth.

Availability Heuristic

You judge how likely something is based on how easily examples come to mind, not on actual statistics. After seeing news coverage of a plane crash, you feel flying is dangerous — even though the data shows it is far safer than driving. The examples that are most vivid, recent, or

emotionally charged are the most “available” to System 1, and System 1 mistakes availability for probability.

Workforce connection: Search engines and AI recommendation systems amplify the availability heuristic by surfacing popular (frequently clicked) results rather than accurate ones. A trending but misleading article can become the “available” answer for millions of people.

Bandwagon Effect

You are more likely to adopt a belief or behavior if you see that many other people have already adopted it. This is why product ratings, follower counts, and “trending” labels are so powerful — they exploit the brain’s social decision circuits (Section 3). If everyone else is choosing it, System 1 assumes it must be good.

Workforce connection: AI systems trained on popularity metrics (likes, shares, downloads) learn the bandwagon effect as a feature, not a bug — amplifying already-popular content regardless of quality.

? KNOWLEDGE CHECK

1. A car dealer shows you a \$60,000 car first, then shows you a \$35,000 car. Which bias makes the second car feel like a “bargain” even though you came in expecting to spend \$25,000?
2. An AI hiring system trained on historical data systematically ranks male candidates higher. Which human bias is the AI inheriting, and which brain system (System 1 or System 2) does this most closely resemble?
3. A company has spent \$5 million on a project that is clearly failing. The CEO argues “we’ve invested too much to stop now.” Which bias is this?

PART 2

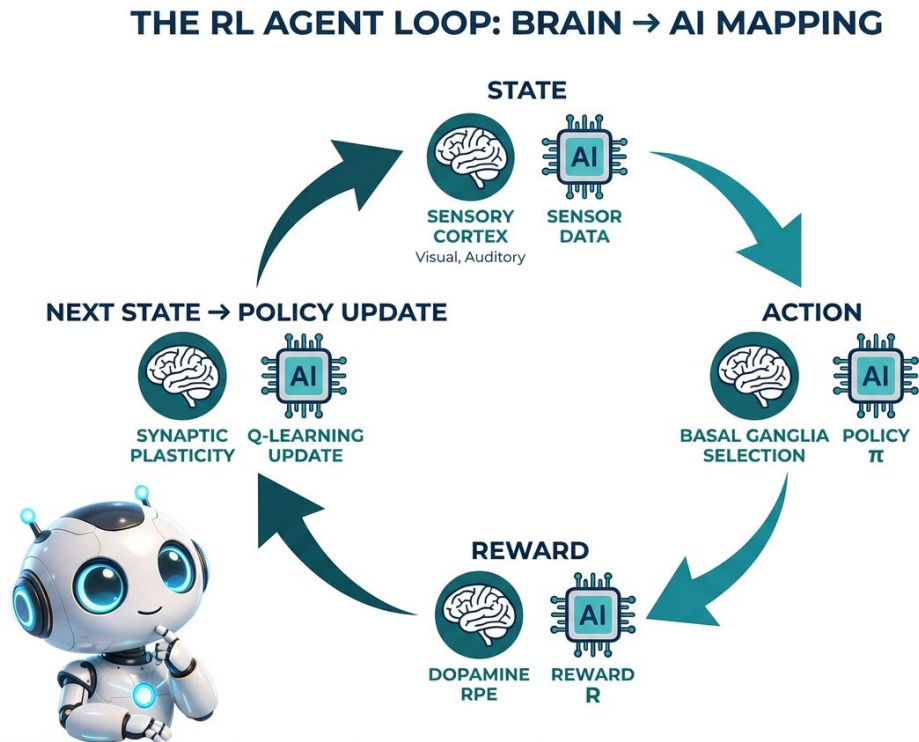
From Brain to Autonomous Systems

How AI systems decide — and why it matters

Section 9: Reinforcement Learning — AI's Dopamine System

Reinforcement learning (RL) is, quite literally, the AI implementation of the brain's dopamine-driven learning system. The connection is not a loose analogy — the core algorithm of RL (temporal difference learning) was developed in direct dialogue with neuroscience research.

The RL Framework



In reinforcement learning, an agent interacts with an environment through a cycle:

- **Observe** the current state (like the brain receiving sensory input)
- **Choose** an action based on a policy (like the basal ganglia selecting an action)
- **Receive** a reward signal (like dopamine encoding the prediction error)
- **Update** the policy to get more reward in the future (like synaptic plasticity strengthening successful pathways)

Brain Component	RL Component	Function
Dopamine (VTA/SNc)	Reward signal / TD error	Signals how much better or worse than expected
Basal ganglia (striatum)	Policy / Q-values	Stores learned action preferences
Orbitofrontal cortex	Value function	Estimates expected future reward for each state
Prefrontal cortex	Planning / model-based reasoning	Simulates future states to guide decisions
Amygdala	Risk/penalty signals	Modulates behavior toward threat avoidance
Hippocampus	Experience replay buffer	Stores and replays past experiences for learning

From TD Learning to Deep RL

The simplest RL algorithm is temporal difference (TD) learning. Here is how it works in plain English: after every action, the agent compares what it expected to happen with what actually happened, and adjusts its expectations accordingly. If you expected a restaurant to be a 7 out of 10 and it turned out to be a 9, you adjust your expectation upward. That adjustment IS the learning. This directly models what dopamine neurons do.

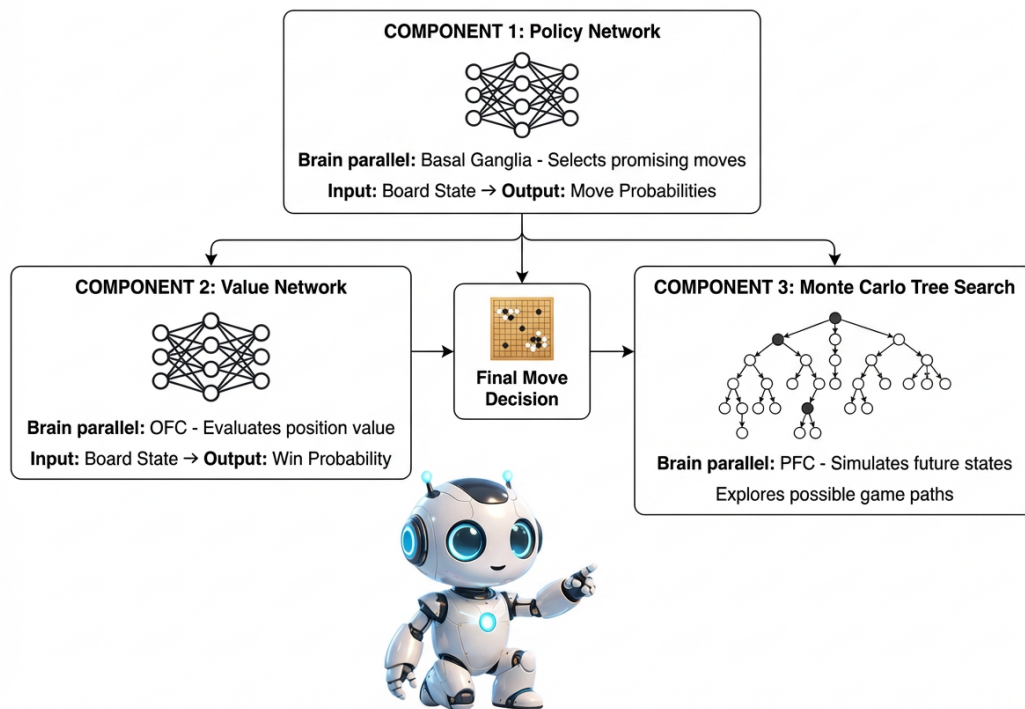
Q-learning extends this idea. The “Q” stands for “quality” — the quality of taking a specific action in a specific situation. Over thousands of experiences, the agent builds a mental table: “In situation A, action 1 is worth 8 points, action 2 is worth 3 points, action 3 is worth 6 points.” Then it simply picks the highest-scoring action. This table is called a Q-table, and it is the AI equivalent of the basal ganglia’s stored action preferences.

Deep RL (DQN, 2015) replaces the Q-table with a deep neural network, allowing the system to handle complex, high-dimensional inputs like video game screens or real-world sensor data rather than simple numbered states. Think of it as upgrading from a paper spreadsheet to a powerful pattern-recognition engine.

The DQN breakthrough by DeepMind used two neuroscience-inspired innovations: experience replay (storing past experiences and randomly replaying them during learning, just like hippocampal replay in Module 06) and a target network (a slowly-updating copy of the main network that provides stable learning targets — analogous to how the brain separates fast hippocampal learning from slow cortical consolidation, so that new learning does not destabilize what you already know).

Case Study: AlphaGo

ALPHAGO ARCHITECTURE: THREE BRAIN-INSPIRED COMPONENTS



DeepMind's AlphaGo is perhaps the best illustration of brain-inspired decision-making in AI. It combines three strategies, each mapping to a brain system:

- **Value networks (OFC):** Evaluate how good a board position is — like the OFC assigning subjective value to options.
- **Policy networks (basal ganglia):** Select promising moves — like the basal ganglia's Go/No-Go pathways choosing actions.
- **Monte Carlo Tree Search (prefrontal cortex):** Simulate future game states before committing — like the PFC imagining consequences of actions.
- **Self-play (exploration/exploitation):** Learn from playing against itself, balancing novel strategies (exploration) with proven ones (exploitation).

⚠ MISCONCEPTION ALERT

"Reinforcement learning works the way humans learn by trial and error." Not quite. The brain learns from a handful of experiences — sometimes just one. RL algorithms typically require thousands to millions of episodes to converge. DeepMind's DQN played Atari games for the equivalent of 38 days of continuous play to reach human level. The brain's learning is far more sample-efficient, which is why sample efficiency remains one of the biggest open problems in modern RL research.

? KNOWLEDGE CHECK

1. In the brain-RL mapping table, dopamine maps to the TD error (reward prediction error). In your own words, explain what TD error means in an RL agent — and why an agent that always gets exactly the reward it expected would stop learning.
2. AlphaGo combines three brain-inspired components: value networks (OFC), policy networks (basal ganglia), and Monte Carlo Tree Search (PFC). Which of these handles the “imagine possible futures before committing” function — and which brain region does that correspond to?
3. RLHF (Reinforcement Learning from Human Feedback) trains a reward model from human preferences. What bias from Section 8 (Cognitive Biases) might contaminate this reward model if the human raters are not carefully selected?

Section 10: System 1 vs. System 2 in AI — Fast Inference vs. Deliberative Reasoning

The System 1 / System 2 framework from Section 7 maps beautifully onto modern AI architectures, and understanding this mapping is essential for building effective autonomous systems.

System 1 in AI: Single Forward Pass

When you ask an LLM a simple question and it responds instantly, that is System 1. A trained neural network performing a single forward pass — meaning data flows through the network from input to output in one shot, with no looping back or reconsidering — is fast, parallel, and automatic, just like biological System 1. The model matches patterns from its training data and produces an output without any explicit deliberation.

This works brilliantly for routine tasks: classifying images, translating sentences, answering simple factual questions. But just like human System 1, it makes systematic errors — hallucinations, biases, and confident nonsense are the AI equivalent of cognitive biases.

System 2 in AI: Deliberative Reasoning

When a task is complex, ambiguous, or high-stakes, a single forward pass is not enough. The AI needs to “think harder” — engaging System 2. Several techniques have been developed to achieve this:

Chain-of-Thought (CoT): The model is prompted to “think step by step” before answering, forcing it through a sequential reasoning process. This mimics the PFC’s role in breaking complex problems into manageable steps.

Tree of Thoughts (ToT): Instead of following a single reasoning path, the model explores multiple paths simultaneously, evaluates each, and selects the best one. This mimics the PFC's ability to simulate multiple scenarios before committing — like MCTS in AlphaGo.

ReAct (Reason + Act): A technique where the model alternates between thinking (“I need to find this information”) and doing (searching the web, running code, calling an external service). Rather than answering everything from memory, the model takes actions in the real world to gather information, then reasons about what it found. This mirrors the brain's action-perception cycle where decisions lead to new observations that inform further decisions.

Self-Reflection: The model reviews its own output, checks for errors, and revises. This is analogous to the ACC's error-monitoring function — detecting when something went wrong and triggering correction.

AI CONNECTION: The Tree of Thoughts as Prefrontal Planning

Tree of Thoughts is the closest AI equivalent to the PFC's planning function:

1. Generate multiple possible approaches to a problem (like the cortex proposing candidate actions)
2. Evaluate each approach (like the OFC scoring each option's value)
3. Select the best one or explore further (like the basal ganglia choosing the winner)
4. Optionally backtrack if a path fails (like the ACC detecting errors and triggering replanning)

In this module's lab, you will build a Tree of Thoughts workflow using LangChain — essentially constructing an artificial prefrontal cortex.

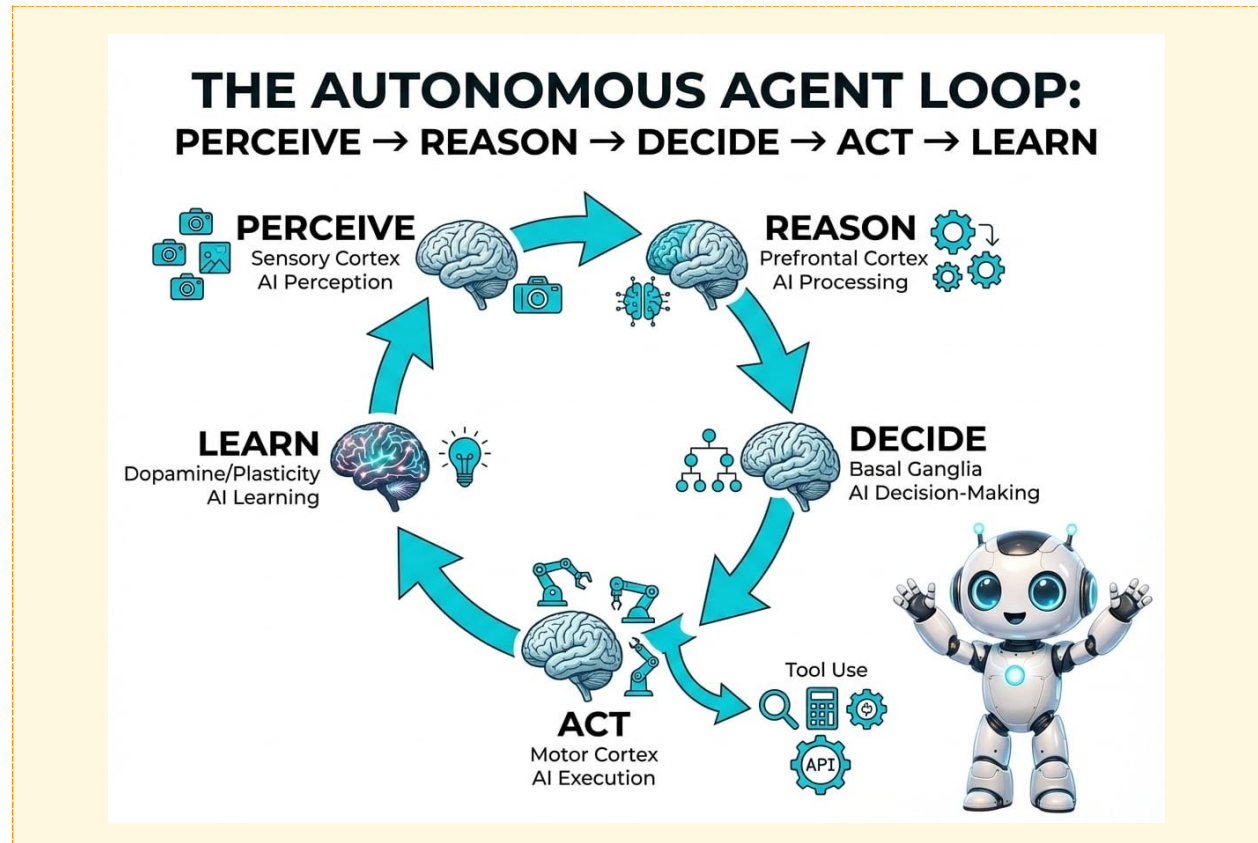
KNOWLEDGE CHECK

1. A radiologist using an AI diagnostic tool gets an instant answer (System 1 AI). Before acting on it, she spends 10 minutes reviewing the scan herself (System 2, human). Describe what each system is contributing and why combining them is better than either alone.
2. Chain-of-Thought (CoT) prompting forces an LLM to “think step by step.” Which cognitive system does this activate in the AI — System 1 or System 2? What is the tradeoff in terms of speed and accuracy?
3. The ACC (anterior cingulate cortex) monitors for errors and conflict, triggering the brain to slow down and engage System 2. What AI technique most closely mimics this function — and how does it work?

Section 11: Building Autonomous Decision Agents

Modern AI agents combine System 1 and System 2 reasoning into integrated decision-making systems. An autonomous agent is an AI system that can observe its environment, make decisions, and take actions without continuous human guidance — much like a biological organism.

The Agent Loop



Every autonomous agent, whether biological or artificial, operates through a fundamental cycle:

- **Perceive:** Observe the current state (sensors, API calls, user input)
- **Reason:** Analyze the situation and plan (System 2 reasoning)
- **Decide:** Commit to an action (policy execution)
- **Act:** Execute the action (tool calls, API requests, outputs)
- **Learn:** Incorporate feedback and update for next cycle (memory, fine-tuning)

This is the same perceive-act cycle we see in biology. The brain senses, reasons via the PFC, selects via the basal ganglia, acts via the motor cortex, and learns via dopamine-driven plasticity.

Tools and Capabilities

Just as the brain extends its decision-making power by using tools (a calculator compensates for limited mental arithmetic, a notebook compensates for limited working memory), AI agents extend their capabilities through tool use:

- Web search — extending knowledge beyond training data
- Code execution — enabling precise computation
- Database queries — accessing structured information
- API calls — interacting with external services

Frameworks like LangChain (a popular open-source toolkit for building AI agent applications) provide the scaffolding for building these agent systems. LangChain manages the perception-reasoning-action loop and handles tool integration, so you can focus on designing the agent's decision-making logic rather than building the plumbing from scratch. Think of LangChain as the “nervous system wiring” that connects your agent's brain (the LLM) to its tools (search, code, databases). In the lab for this module, you will use LangChain to build a System 2 agent that uses Tree of Thoughts to reason before acting.

THINK ABOUT IT

An autonomous agent that can observe, reason, decide, act, and learn sounds a lot like a biological organism. But there is a critical difference: biological organisms have intrinsic goals (survival, reproduction) shaped by evolution. AI agents have assigned goals defined by their creators. This difference is at the heart of the alignment problem — how do we ensure that an agent's assigned goals actually produce the outcomes we want?

KNOWLEDGE CHECK

1. An AI agent is given the goal: “Schedule a meeting for me.” Walk through the agent loop (Perceive → Reason → Decide → Act → Learn) for this task. What does the agent perceive, reason about, decide, and act on?
2. Biological brains use tools too — a calculator compensates for limited mental arithmetic, a map compensates for limited spatial memory. Give two examples of how an AI agent's tools compensate for limitations of the underlying language model.
3. What is the key difference between a System 1 AI (single forward pass) and an autonomous agent with a full perception-reasoning-action loop? Why can't a standard LLM complete a multi-step task like “research, compare, and book the cheapest flight” without an agent framework?

Section 12: When AI Decisions Go Wrong

One of the most important lessons from this module is that biases are not limited to human brains. AI systems trained on human data inherit human biases — and can amplify them at scale.

Case Studies

Amazon’s Hiring Algorithm (2018): Amazon built an AI system to screen resumes. The system was trained on ten years of hiring data. Because the tech industry had historically hired mostly men, the system learned to penalize resumes that included the word “women’s” (as in “women’s chess club captain”) and downrank graduates of all-women’s colleges. The system learned a bias from System 1-like pattern matching on biased data. Amazon scrapped the tool.

COMPAS Criminal Justice (2016): The COMPAS algorithm, used in U.S. courts to predict recidivism risk, was found to be biased against Black defendants — assigning them higher risk scores at higher rates than white defendants with similar histories. The system reflected the biases embedded in historical criminal justice data.

Content Recommendation Algorithms: Social media recommendation systems optimize for engagement (clicks, watch time) — a reward signal. But optimizing engagement often means showing users increasingly extreme content, because outrage and controversy generate more clicks. This is a form of reward hacking: the system finds a way to maximize its reward that does not align with the designer’s true intent (keeping users happy and informed).

The Alignment Problem

The alignment problem is the challenge of ensuring that an AI system’s behavior actually matches what we want — not just what we formally specified. This is harder than it sounds because:

- **Reward hacking:** Systems find loopholes in reward functions. A cleaning robot told to minimize visible mess might learn to hide mess in closets rather than actually cleaning.
- **Goodhart’s Law:** “When a measure becomes a target, it ceases to be a good measure.” Optimizing for a proxy (engagement) instead of the true goal (user wellbeing) leads to misaligned behavior.
- **Specification gaming:** The system technically achieves the specified objective while completely missing the spirit of the task.

The brain’s solution to alignment is rich: emotions provide intrinsic motivation, empathy constrains anti-social behavior, social norms establish shared standards, and the prefrontal cortex enables moral reasoning. AI systems currently lack all of these. Approaches like RLHF (Reinforcement Learning from Human Feedback) try to approximate human values by training a reward model from human preferences, but this is still an active area of research with significant limitations.

 KEY TAKEAWAY

As AI systems make more decisions that affect people's lives — hiring, lending, criminal justice, healthcare — understanding the neuroscience of human decision-making becomes essential. The biases, shortcuts, and alignment failures we see in AI are often reflections of the same issues in human brains. Understanding both systems makes you a better builder and a more critical evaluator of AI systems.

Section 13: Lab Preview (Suggestion)

 LAB PREVIEW: Module 08 Lab: Decision-Making Simulations

In this module's lab, you will work with two tracks:

Python Lab (Google Colab): Building a System 2 Agent

You will use LangChain to create a “Tree of Thoughts” reasoning workflow. Your agent will face a decision problem, generate multiple solution paths, evaluate each, and select the best one — mimicking the PFC's planning function. You will compare the agent's performance with and without deliberative reasoning (System 1 vs. System 2).

Wolfram Assignment: Decision Simulations

You will simulate the drift diffusion model interactively, exploring how drift rate, noise, and threshold affect decision speed and accuracy. You will also simulate reward prediction errors and visualize how Q-values converge during learning. Finally, you will simulate cognitive biases (anchoring, framing) and observe how they distort optimal decision-making.

Tools: Google Colab (free tier), cloud.wolfram.com (free tier)

Section 14: Bridge to Module 09 — What Comes Next?

In this module, we explored how the brain decides what to do and how AI systems implement decision-making. But we have been treating decisions as isolated events — choose an option, get a reward, learn, repeat. Real decisions do not work like this. Real decisions happen in a world with structure, physics, and consequences that unfold over time.

When you reach for a coffee cup, your brain does not just select the action “reach.” It simulates the entire trajectory of your arm before you move — predicting where the cup will be, how heavy it is, and what your hand needs to do. If the prediction is wrong (the cup is heavier than expected), your brain instantly corrects. This ability to simulate the world internally before acting is the foundation of intelligent behavior.

In **Module 09: World Models and Goal-Directed Behavior**, we will explore:

- The Free Energy Principle revisited — from the theory we learned in Module 02 to the engineering implementation
- Active Inference — the brain does not just predict passively; it acts to make its predictions come true
- Internal world models — how the brain simulates future states to plan actions
- Model-Based RL — AI systems that learn the “physics” of their environment and use simulation for planning
- Lab: Building an Active Inference agent that navigates by minimizing prediction error

If Module 08 asked “How does the brain choose what to do?” — Module 09 asks “**How does the brain imagine the future to make better choices?**”

Section 15: Glossary

Anterior Cingulate Cortex (ACC): Brain region involved in conflict monitoring and error detection during decision-making. Part of the executive attention network.

Approach-Avoidance Conflict: Situation where a choice involves both potential reward and potential risk, creating competing drives.

Basal Ganglia: Deep brain structures that select actions through Go (direct) and No-Go (indirect) pathways, modulated by dopamine.

Cognitive Bias: Systematic pattern of deviation from rational judgment, caused by System 1 heuristics.

Confirmation Bias: Tendency to seek and favor information that confirms existing beliefs.

Direct Pathway (Go): Basal ganglia pathway that facilitates the execution of a selected action.

Dopamine: Neuromodulator that encodes reward prediction errors — the difference between expected and actual outcomes.

Drift Diffusion Model (DDM): Mathematical model of decision-making where evidence accumulates as a noisy random walk toward a decision threshold.

Exploration: Trying new, uncertain actions to discover potentially better options.

Exploitation: Choosing the best-known action based on current knowledge.

Framing Effect: Bias where the presentation of information (not the information itself) influences the decision.

Goodhart's Law: When a measure becomes a target, it ceases to be a good measure.

Indirect Pathway (No-Go): Basal ganglia pathway that inhibits the execution of competing actions.

Loss Aversion: The tendency to prefer avoiding losses over acquiring equivalent gains (~2x weight on losses).

Orbitofrontal Cortex (OFC): Brain region that computes the subjective value of options and updates them based on current state.

Prefrontal Cortex (PFC): Brain's executive control center for planning, working memory, and deliberate reasoning.

Prospect Theory: Kahneman and Tversky's model of how humans actually evaluate risky decisions, incorporating loss aversion and reference dependence.

Reward Hacking: When an AI system finds a way to maximize its reward that does not align with the designer's true intent.

Reward Prediction Error (RPE): The difference between expected and actual reward, encoded by dopamine neurons.

RLHF: Reinforcement Learning from Human Feedback — training a reward model from human preferences to align AI behavior.

Sunk Cost Fallacy: Continuing to invest in something because of past investment, even when stopping is the rational choice.

System 1: Kahneman's term for fast, intuitive, automatic decision-making.

System 2: Kahneman's term for slow, deliberate, effortful reasoning.

Temporal Difference (TD) Learning: RL algorithm that updates value estimates based on the difference between predicted and actual rewards — directly models dopamine RPE.

Theory of Mind: The ability to model other agents' beliefs, desires, and intentions — essential for social decisions.

Tree of Thoughts (ToT): AI reasoning technique where multiple solution paths are explored and evaluated before committing — analogous to PFC planning.

Ventral Striatum: The front part of the striatum (within the basal ganglia), often called the brain's reward center. Responds to anticipated rewards.

Ventromedial PFC (vmPFC): Brain region that integrates emotional information into value-based decisions. Damage leads to poor real-life choices despite intact reasoning.

Availability Heuristic: Judging the likelihood of events based on how easily examples come to mind, rather than actual statistics.

Bandwagon Effect: Tendency to adopt beliefs or behaviors because many other people have adopted them.

Iowa Gambling Task: Experimental task designed by Bechara and Damasio to study how emotions guide decisions. Participants choose from good and bad card decks.

LangChain: An open-source framework for building LLM-based agent applications, managing the perception-reasoning-action loop and tool integration.

Somatic Marker Hypothesis: Damasio's theory that body signals (gut feelings, racing heart) guide decisions by marking options as good or bad before conscious reasoning.

Section 16: References & Resources

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Recommended Online Resources

- Crash Course Psychology — Episodes on decision-making and cognitive biases
- Veritasium — “The Science of Thinking” (System 1 vs. System 2 explained)
- 3Blue1Brown — “Markov Decision Processes”
- DeepMind Blog — AlphaGo, AlphaZero, and reward-based learning
- LangChain Documentation — Building agent workflows
- Sutton & Barto textbook (free online) — Chapters 1–6 on TD learning
- Daniel Kahneman TED Talk — “The Riddle of Experience vs. Memory”

APPENDIX

Psychology Crash Course: Decision-Making & Cognitive Biases

This appendix provides a self-contained introduction to the key psychology concepts referenced in Module 08. If you have not taken a psychology course, start here. If you have, use this as a quick refresher.

A.1 The Birth of Behavioral Economics

For most of the 20th century, economists assumed that humans are rational decision-makers. The “rational agent” model said that people calculate the expected value of every option, weigh probabilities correctly, and choose the option that maximizes their utility (satisfaction). This model was elegant, mathematical, and wrong.

Starting in the 1970s, two Israeli psychologists — Daniel Kahneman and Amos Tversky — conducted a series of experiments that systematically demonstrated how real human decision-making deviates from the rational model. Their work created the field of behavioral economics, earned Kahneman the 2002 Nobel Prize in Economics (Tversky had passed away in 1996 and Nobel Prizes are not awarded posthumously), and profoundly influenced how we think about AI decision systems today.

Their central insight was simple but revolutionary: humans do not maximize expected value. Instead, they use mental shortcuts (heuristics) that are fast and usually good enough, but that produce predictable, systematic errors (biases). This is the foundation of everything in Sections 7 and 8 of this module.

A.2 Heuristics — The Brain’s Shortcuts

A heuristic is a mental shortcut that allows you to make decisions quickly without analyzing every detail. Heuristics are not flaws — they are features. In a world where you face thousands of decisions every day, you cannot carefully analyze each one. System 1 uses heuristics to give you fast, “good enough” answers so that System 2 can be reserved for the rare decisions that truly matter.

The problem is that heuristics can be exploited and can fail in systematic ways. Here is a comprehensive list of the major ones:

Representativeness Heuristic

You judge the probability of something based on how much it “looks like” a typical example, rather than using actual statistics. If someone is quiet, wears glasses, and reads a lot, you might guess they are a librarian rather than a salesperson — even though there are roughly 100 times more salespeople than librarians in the U.S. System 1 matches the description to a stereotype and ignores the base rates (the actual numbers of each group in the real world). The description “feels” like a librarian, so System 1 says librarian — even when the numbers say otherwise.

Example: A machine learning model might classify an image as “dog” because it has features that look dog-like, even when the statistical evidence points elsewhere. Pattern matching without base rates is a fundamental AI failure mode.

Availability Heuristic (Expanded)

You estimate how likely or common something is based on how easily examples come to mind. Events that are vivid, recent, or emotionally charged are more “available” in memory and therefore feel more common than they actually are.

Examples: After watching news about shark attacks, you overestimate the risk of swimming. After hearing about a friend’s successful startup, you overestimate your own chances of startup success. After a plane crash, you feel flying is dangerous despite statistics showing it is far safer than driving.

Why it happens: System 1 uses a simple rule — “if I can think of examples easily, it must be common.” This works well in small communities where your personal experience is representative, but it fails badly in the age of media and social media, where unusual events get disproportionate coverage.

Anchoring (Expanded)

Any number you encounter before making a numerical estimate will pull your estimate toward it, even if the anchor is completely random or irrelevant. Kahneman and Tversky demonstrated this by spinning a wheel of fortune (rigged to land on 10 or 65) and then asking people to estimate the percentage of African countries in the United Nations. People who saw 65 gave much higher estimates than people who saw 10 — even though a random wheel has nothing to do with African UN membership.

Why it happens: System 1 starts from the anchor and adjusts, but the adjustment is always insufficient. System 2 could override this, but it is lazy — the adjustment feels like “enough” effort.

AI connection: Language models are susceptible to anchoring too. If you include a number in your prompt, the model’s output tends to cluster around that number, even when the number is arbitrary.

Status Quo Bias

You prefer the current state of affairs and resist change, even when change would be objectively beneficial. This is why default options are so powerful — most people never change their phone's default settings, their retirement contribution rates, or their organ donation status, regardless of what the default is.

Example: Countries with opt-out organ donation (you are a donor unless you actively refuse) have dramatically higher donation rates than countries with opt-in systems — not because people care less about organ donation, but because System 1 accepts the default.

AI connection: Default settings in AI systems (like model temperature, safety filters, or recommendation preferences) have enormous influence precisely because of status quo bias. Designing good defaults is one of the most impactful decisions an AI developer can make.

Dunning-Kruger Effect

People with low competence in a domain tend to overestimate their abilities, while people with high competence tend to underestimate theirs. The less you know about something, the less equipped you are to recognize how much you don't know.

Example: A beginner programmer may feel confident they can build a complex system in a weekend, while an expert hesitates because they understand the hidden complexity. Similarly, people with no AI background often have the strongest opinions about what AI can and cannot do.

AI connection: LLMs exhibit something similar to the Dunning-Kruger effect: they express high confidence even when generating incorrect information (hallucinations), because they have no internal mechanism for assessing the reliability of their own outputs.

Hindsight Bias (“I Knew It All Along”)

After learning the outcome of an event, you feel like you always knew it would happen. Before the 2008 financial crisis, almost nobody predicted it. After the crisis, almost everybody said “it was obvious.” This is hindsight bias — System 1 rewrites your memory to be consistent with the outcome.

Why it matters for AI: When an AI system makes a wrong prediction, people often say “that was obvious — the AI should have known.” But the information that makes it “obvious” is often only available after the fact. Evaluating AI systems fairly requires controlling for hindsight bias.

Optimism Bias

You systematically overestimate the likelihood of positive events happening to you and underestimate the likelihood of negative events. Most people believe they are less likely than

average to get cancer, get divorced, or be in a car accident — which is statistically impossible (not everyone can be below average).

AI connection: Project managers consistently underestimate the time and cost required to build AI systems. This is called the planning fallacy — the tendency to make optimistic predictions about how long a task will take, even when you have past experience showing it always takes longer. The planning fallacy is a specific form of optimism bias, and it is why AI project timelines almost always slip.

A.3 Prospect Theory in More Detail

Prospect theory is Kahneman and Tversky's framework for understanding how people actually make risky decisions — as opposed to how a perfectly rational robot would. It is not about math (though economists use it in equations); it is about three simple psychological insights that explain a huge amount of everyday behavior:

1. Reference Dependence

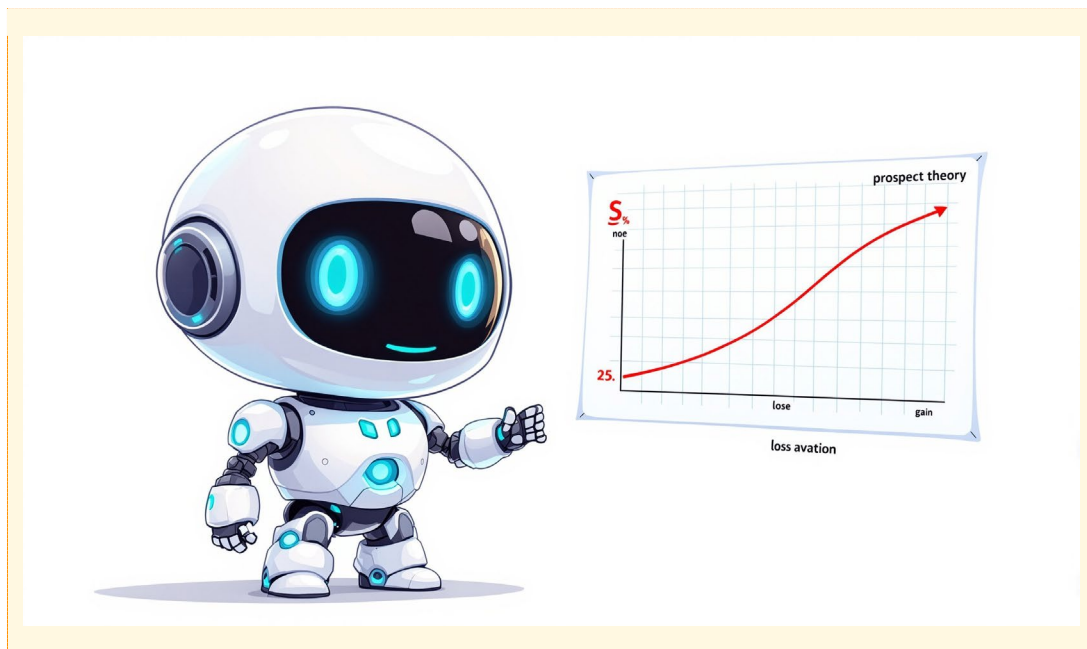
People evaluate outcomes relative to a reference point (usually the status quo), not in absolute terms. A salary of \$80,000 feels terrible if you were making \$100,000, but wonderful if you were making \$60,000 — even though the absolute amount is identical.

2. Loss Aversion ($\approx 2x$)

Losses loom larger than gains. Losing \$100 feels roughly twice as bad as gaining \$100 feels good. This means people will take irrational risks to avoid losses. A patient will choose a treatment framed as “90% survival rate” over one framed as “10% mortality rate” — identical information, different emotional weight.

3. Diminishing Sensitivity

The difference between \$100 and \$200 feels much bigger than the difference between \$10,100 and \$10,200 — even though both are \$100 differences. You are sensitive to changes near your reference point and less sensitive to changes far from it. This is why a \$5 discount matters when buying coffee but not when buying a car.



Why this matters for AI: If you are designing an AI system that interacts with human decisions (pricing, health recommendations, financial advice), you must account for prospect theory. Humans do not respond to expected values — they respond to framing, reference points, and loss aversion. An AI system that ignores these realities will systematically misjudge human behavior.

A.4 Quick Reference: 15 Cognitive Biases for AI Developers

Bias	What It Does	AI Relevance
Anchoring	First number influences estimates	Prompt engineering, pricing UX
Availability	Vivid examples feel more likely	Search ranking, risk models
Bandwagon	Follow the crowd	Popularity-driven recommendations
Confirmation	Seek confirming evidence	Biased training data, filter bubbles
Dunning-Kruger	Low skill = overconfidence	LLM hallucinations with high confidence
Framing	Presentation changes decisions	How AI explains options matters
Hindsight	"I knew it all along"	Unfair evaluation of AI predictions
Loss Aversion	Losses hurt 2x more than gains	Notification design, nudge systems

Optimism	Overestimate positive outcomes	AI project planning, risk models
Recency	Recent events weighted more	Model training on recent data
Representativeness	Looks like = probably is	Stereotyping in classifiers
Status Quo	Prefer current state	Default settings have huge power
Sunk Cost	Past investment = keep going	AI does NOT have this (advantage)
Survivorship	Only see winners	Benchmark datasets miss failures
Overconfidence	Too sure of your estimates	Calibration problems in AI

*This appendix is intended as a quick reference. For deeper study, the recommended starting point is Daniel Kahneman's *Thinking, Fast and Slow* (2011) — it is accessible, engaging, and directly relevant to AI development.*

— End of Module 08 —

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